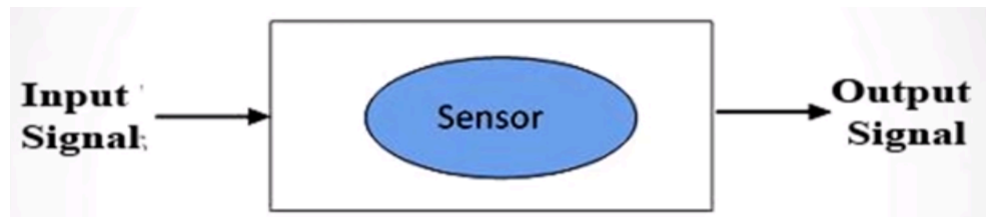


Sensors

What is a Sensor?

A sensor is a device that:

- Detects changes in physical quantities such as:
 - Temperature
 - Light
 - Sound
 - Pressure
 - Motion
 - Humidity
 - Flow
- Converts these changes into a readable form (usually still physical or a simple variation like resistance).



Key Points about Sensors

- Input and output are both physical (non-electrical) in nature.
- Sensors are highly accurate when properly calibrated.
- Sensors usually produce low-level signals, so calibration is needed.
- They do not necessarily convert energy into another form.

Example

- **A thermistor:**
 - Senses temperature change.
 - Changes its resistance.

- Does not directly convert it into an electrical signal usable by a processor.

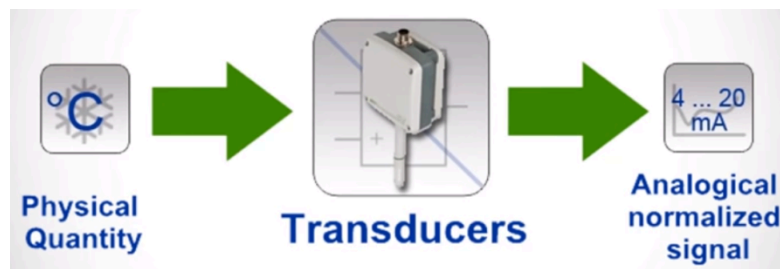
What is a Transducer?

A transducer is a device that:

- Converts energy from one form to another.
- Typically converts non-electrical signals into electrical signals.

Types of Energy a Transducer Can Handle

- Mechanical
- Thermal
- Magnetic
- Electrical
- Chemical
- Radiation



A sensor becomes part of a transducer when combined with additional circuitry.

Example

- Thermistor alone → Sensor
- Thermistor + circuit (e.g., thermometer) → Transducer

Difference Between Sensor and Transducer

The words sensor and transducer are often used interchangeably, but they are actually different devices with different roles.

Main Difference

- A sensor detects or senses a change in the environment and gives output in the same form (usually physical).
- A transducer converts a measurement from one form of energy to another, usually into an electrical signal. This conversion process is called transduction.

Although many sensors are part of transducers, not all transducers are sensors.

Comparison Between Sensor and Transducer

Feature	Sensor	Transducer
Working Principle	A sensor detects or senses a physical quantity and makes it readable, but keeps the output in the same form (non-electrical/physical)	A transducer detects a physical quantity and converts it from one form to another, usually non-electrical to electrical
Function	Only sensing/detecting	Sensing + conversion (transduction)
Output Type	Output remains in same physical form	Output is usually converted into electrical signal

Examples	Thermistor, motion sensor, pressure switch	Microphone, pressure transducer, linear transducer
Uses / Applications	- Patient monitoring systems - Infrared toilet flush systems - Liquid dispensing in drink machines	- HVAC monitoring systems - Engine control systems - Steering systems - Ramp and bridge lifting systems

Types of Transducers

1. Input Transducer

- Converts physical (non-electrical) quantity → electrical signal
- Example:
 - Linear transducer converts position → electrical signal

2. Output Transducer (Actuator)

- Converts electrical signal → physical output
- Example:
 - Light bulb converts electricity → light

Measuring System Structure

A complete measuring system has three main parts:

1. Sensor

- Detects physical change
- Output is still physical (non-electrical)

2. Signal Conditioning Unit

- Converts sensor output into usable electrical signals
- Makes it suitable for processing

Examples:

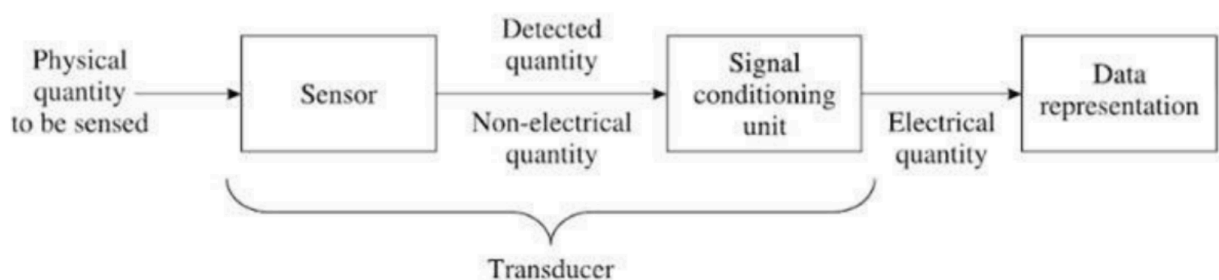
- Analog to Digital Converter (ADC)
- Amplifier
- Filter
- Rectifier
- Modulator

3. Data Representation Device

- Displays or records the output

Examples:

- Scale
- LCD display
- Signal recorder



Example (Strain Gauge)

- A strain gauge senses strain → changes resistance (sensor).
- A Wheatstone bridge circuit converts resistance → voltage.

- Together:
 - $\text{Sensor} + \text{Signal Conditioning} = \text{Transducer}$

Physical Quantities and Electrical Conversion

Sensors detect physical quantities such as:

- Temperature
- Pressure
- Force
- Motion
- Displacement
- Humidity
- Light
- Flow

These are converted into electrical properties like:

- Resistance
- Inductance
- Capacitance

Then further converted into:

- Voltage
- Current

Hardware Calibration

Sensors usually produce very low-level signals, which are not directly suitable for a processor or microcontroller. So, calibration is required to convert these signals into an usable and readable form.

Types of Input to Processor

1. Digital Input

- Works with only two levels:
 - 0 → Ground (0V)
 - 1 → Greater than 3.5V
- Used for simple ON/OFF type signals

2. Analog Input

- Works with a range of voltages
- Typical range:
 - 0V to 5V
- Represents continuous values (e.g., temperature, pressure)

Conversion Process

Since processors understand digital values, analog signals must be converted.

Process:

Analog Voltage → ADC (Analog to Digital Converter) → Numerical Value

Example:

- For an 8-bit ADC:
 - Output range = 0 to 255
 - 0V → 0
 - 5V → 255

Why Calibration is Needed

- Sensor outputs are small (low-level signals)

- These signals are not directly usable by processors
- Calibration:
 - Adjusts the signal into a readable and usable range
 - Ensures accurate measurement and processing

For efficient and correct results:

- Lower input value → mapped to lower output range
- Higher input value → mapped to higher output range

This ensures:

- Proper scaling
- Better accuracy
- Reliable system performance

Classification of Sensors

Sensors can be classified in two main ways:

1. By Material
2. By Engineering / Application

1. Sensors by Material

These sensors use specific materials that naturally react to environmental changes. Because of this, they do not need complex extra circuits to produce a signal.

When the material senses a change, it directly produces a measurable output (often voltage, resistance, or charge change).

Types of Material-Based Sensors

- **Thermistor**
 - Sensitive to temperature
 - Resistance changes with temperature
- **Piezoelectric Material**
 - Generates voltage when pressure or vibration is applied
- **Pyroelectric Material (IR)**
 - Detects infrared radiation (heat)
 - Produces electrical signal when temperature changes
- **Electromagnetic**
 - Works based on magnetic field changes
- **Semiconductor**
 - Properties change with light, temperature, or voltage
- **Radioactive Material**
 - Used to detect radiation
- **Capacitive**
 - Detects changes in capacitance (distance, humidity, touch)
- **Resistive**
 - Detects changes in resistance (e.g., strain gauge)
- **Conductive**
 - Detects changes in conductivity (e.g., liquid level)

Key Idea (Material-Based Sensors)

- Material itself does the sensing
- Produces direct electrical variation
- Minimal external circuitry needed

2. Sensors by Engineering / Application

These sensors are classified based on their design, function, and real-world usage.

Types of Engineering-Based Sensors

- **Gyroscope**
 - Measures orientation and angular velocity
- **Microphone**
 - Converts sound waves into electrical signals
- **Camera**
 - Captures images using light sensors
- **LiDAR (Light Detection and Ranging)**
 - Measures distance using laser light
- **Pulse Oximeter**
 - Measures oxygen level in blood
- **Level Sensor**
 - Detects level of liquids or materials
- **Hall-effect Sensor**
 - Detects magnetic fields
- **Break-Beam Sensor**
 - Detects object interruption using light beam
- **Vibration Sensor**
 - Detects vibration or movement
- **Accelerometer**
 - Measures acceleration and motion

Key Idea (Engineering-Based Sensors)

- Classified based on function and application
- Often combine multiple materials and circuits
- Widely used in real-world systems

Note

- Material-based sensors depend on the natural properties of materials
- Engineering-based sensors depend on design and application
- Both types ultimately help in detecting physical changes and producing signals for processing

Classification of Sensors (Based on Operation)

Sensors are classified into two types:

- Active Sensors
- Passive Sensors

1. Active Sensors

Active sensors provide their own signal and use the interaction of that signal with the environment to measure physical properties.

Characteristics

- Require an external power source
- Generate (transmit) energy and also detect it
- Consist of two main parts:
 - Emitter → sends signal
 - Detector → receives signal
- Work by analyzing how the emitted signal changes after interacting with the environment

Working Idea

- Emit energy (light, sound, electromagnetic waves)
- Receive the reflected or modified signal
- Process the changes to measure the environment

2. Passive Sensors

Passive sensors measure physical properties directly from the environment without generating their own signal.

Characteristics

- Do not require an external power source to produce the sensing signal

- Do not transmit energy
- Only detect energy coming from an external source

Working Idea

- Detect naturally available energy (heat, light, pressure, etc.)
- Convert it into a measurable output

Key Differences Between Active and Passive Sensors

Feature	Active Sensor	Passive Sensor
Energy Source	Requires external power source	Does not require external power for sensing
Signal Generation	Generates its own signal	Does not generate signal
Operation	Transmits and detects energy	Only detects existing energy
Components	Has emitter + detector	Only detector
Working Principle	Sends signal and measures its interaction with environment	Measures natural physical changes from environment
Complexity	More complex	More simple
Accuracy / Control	More controlled and precise (can work in the dark, etc.)	Depends on environmental conditions
Examples	LiDAR, Radar, Ultrasonic sensor	Thermistor, Light sensor, IR sensor

Light-Based Sensors

Types

- Camera
- Infrared (IR) Sensor
- LDR (Light Dependent Resistor)
- X-ray Sensor
- UV Sensor
- Heartbeat Sensor
- Laser Sensor

1.Camera Sensor

How to Identify

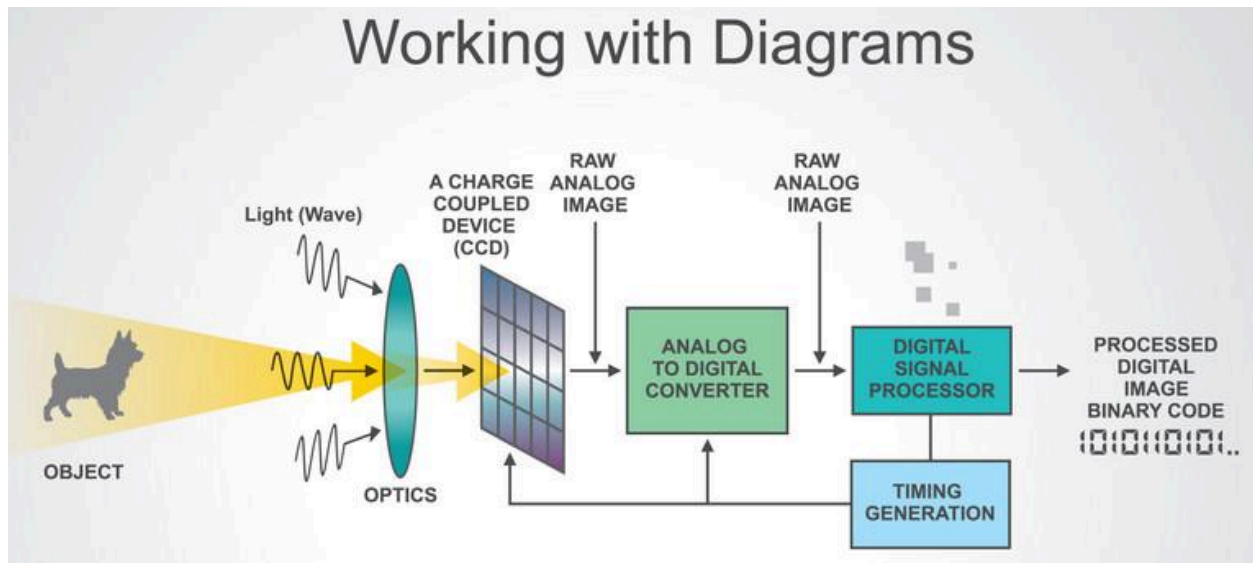
A camera sensor is identified in scenarios mentioning:

- Image capturing
- Video recording
- Pixels or resolution
- Lens or optics
- Image processing

Working Mechanism

- An object reflects light waves toward the camera
- Light enters through the optics (lens)
- The lens focuses light onto the sensor, typically a **CCD (Charge Coupled Device)**
- The CCD converts photons (light) → electrical signals
- This produces a raw analog image
- The analog signal is sent to an Analog-to-Digital Converter (ADC)
- ADC converts analog signal → digital values
- The digital data is processed by a Digital Signal Processor (DSP):

- Color reconstruction
- Brightness and contrast adjustment
- Sharpening
- Compression (e.g., JPEG format)
- Final output is a processed digital image in binary form (0s and 1s)
- A timing generation unit ensures synchronization between CCD, ADC, and DSP



Example

- Smartphone camera
- CCTV camera
- Webcam
- Digital camera

2. Infrared Sensor

How to Identify

An IR sensor is identified in scenarios involving:

- Heat detection

- Motion sensing
- Invisible radiation detection
- Remote control systems

Working Mechanism

- IR sensors work based on the fact that all objects emit infrared radiation (heat), which is invisible to the human eye.

Active IR Sensor

- Contains: IR LED (Emitter), Photodiode (Detector)
- IR LED sends infrared light
- Light reflects from object
- Photodiode receives reflected light
- When light is received: Resistance changes, Voltage changes
- Some sensors use pyroelectric material
- It produces small voltage when heated or cooled

Passive Infrared (PIR) Sensor

- Does not emit infrared radiation
- Only detects infrared energy from surroundings

How to Identify

A PIR sensor is identified in scenarios involving:

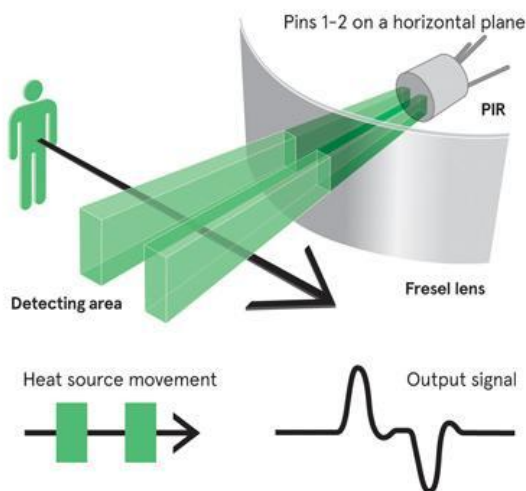
- Motion detection of humans or animals
- Security systems
- Automatic lighting systems

It can also be identified physically by:

- White dome-shaped Fresnel lens
- Three-pin interface (VCC, Ground, Output)
- Adjustable knobs for sensitivity and delay

Working Mechanism

- PIR sensor works using a **pyroelectric sensor and a Fresnel lens**
- The Fresnel lens focuses infrared radiation onto the sensor
- It also divides the detection area into multiple zones
- The internal sensor has two halves
- Under normal conditions:
 - Both halves detect equal ambient heat
 - Output remains stable
- When a warm object (human/animal) moves across the detection area:
 - One half detects the change in heat before the other
 - This creates a voltage difference between the two halves
- This voltage difference triggers the output to go HIGH
- HIGH output indicates **motion detection**



Characteristics

- PIR sensor detects only changes in heat, not constant heat

- It requires a short stabilization time to adjust to room temperature

Final Output

- A digital HIGH signal when motion is detected

Infrared Camera (Thermal Camera)

How to Identify

An infrared camera is identified in scenarios mentioning:

- Heat maps
- Temperature visualization
- Thermal imaging
- Heat signature detection



Working Mechanism

- All objects emit infrared heat energy
- The camera collects this heat using an optical system
- Heat energy falls on a sensor array with many pixels
- Each pixel detects infrared radiation
- Pixels generate electrical signals
- The processor converts signals into temperature values
- Different temperatures are shown using different colors
- Final output is a thermal image showing heat distribution
- The image is displayed and stored in memory

Final Output

- A thermal image representing heat distribution of objects in the scene

Example:

- Detecting fever or high body temperature
- Monitoring inflammation or abnormal blood flow
- Screening patients in hospitals or airports

Infrared in Temperature Sensor

How to Identify

- Measuring temperature without contact
- Industrial temperature monitoring
- Thermal measurement of objects
- “Non-contact thermometer” or “IR thermometer”



Working Mechanism

- Every object emits infrared radiation (heat energy) depending on its temperature
- The IR temperature sensor detects this emitted radiation using an IR detector (thermopile/pyroelectric sensor)
- The sensor converts IR energy into an electrical signal (voltage/current)
- This signal is processed and converted into a temperature value ($^{\circ}\text{C}$ or $^{\circ}\text{F}$)

Output

- A single numerical temperature value

Example

- Digital IR thermometer
- Industrial temperature guns

Infrared in TV Remote (Active IR System)

How to Identify

- Remote control operation
- TV/AC switching
- Short-distance communication
- Line-of-sight requirement

Working Mechanism

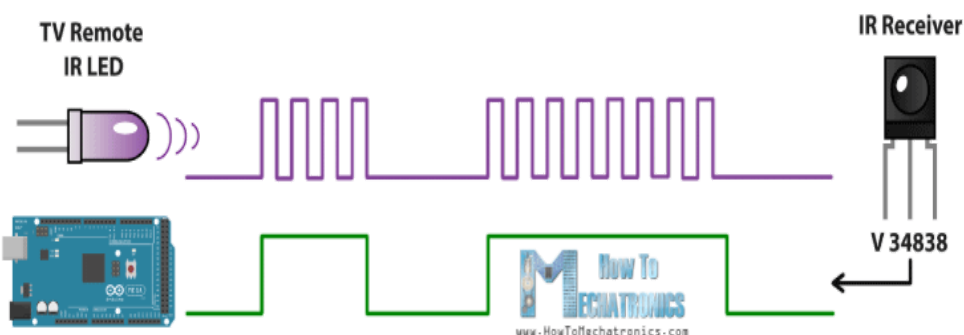
- The remote contains an IR LED (emitter)
- When a button is pressed, it sends encoded IR light pulses
- These pulses carry commands (volume up, channel change, power on/off)
- The TV has an IR receiver (photodiode sensor)
- The receiver detects the pulses and converts them into electrical signals
- The system decodes the signal and performs the required action

Output

- Control commands (not measurement)

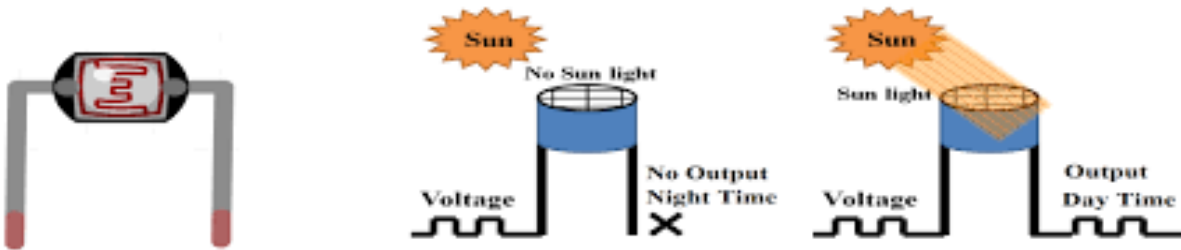
Example

- TV remote
- AC remote
- Set-top box remote



Light Dependent Resistor (LDR)

- A Light Dependent Resistor (LDR) or photo resistor is a device whose resistivity depends on incident electromagnetic radiation (light).
- It is a light-sensitive device.
- It is also called:
 - Photo conductor
 - Photo conductive cell
 - Photocell
- It is made using photo-sensitive semiconductor material.



Working Principle

- LDR resistance changes according to light intensity

When light falls on LDR:

- More free electrons are produced
- Conductivity increases
- Resistance decreases

In darkness:

- Fewer free electrons are present
- Conductivity decreases
- Resistance increases

Key Behavior

- Bright light → low resistance (high conductivity)
- Dark condition → high resistance (low conductivity)

Output Response

- The change in resistance is used in electronic circuits
- It is converted into a voltage or current signal for measurement and control applications

Note

- LDR works by converting light intensity changes into resistance changes, using photo-sensitive semiconductor material.

Example

- automatic street light system

UV Sensor Module

How to Identify

A UV sensor module is identified in scenarios involving:

- Sunlight intensity measurement
- UV index detection
- Skin protection / sunscreen monitoring
- Environmental radiation sensing
- Weather or outdoor monitoring systems

Working Principle

- UV sensor detects ultraviolet (UV) light
- Made of UV-sensitive semiconductor material

Working Mechanism

1. UV Detection

- UV light falls on the sensor
- Sensor absorbs UV energy

2. Electron Generation

- UV energy creates free electrons and holes
- Higher UV intensity produces more charge carriers

3. Electrical Signal Production

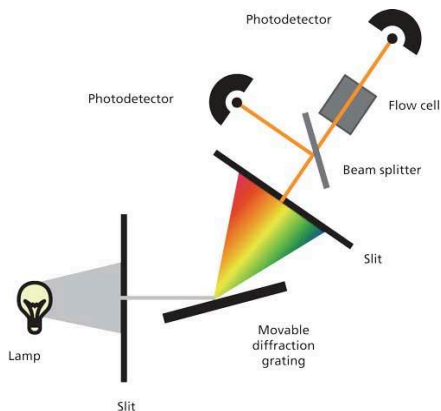
- Conductivity changes
- Voltage/current/resistance changes

4. Signal Processing

- Signal is processed by the circuit
- Converted into readable output proportional to UV intensity

Example

- UV index meters
- Smart weather stations
- Skin protection devices



Output

- A voltage or digital signal representing UV intensity level
- Often displayed as:
 - UV index value
 - Radiation intensity level

Key Point

- More UV exposure → more electrical output
- Less UV exposure → lower output

Note

- UV sensor works by converting ultraviolet radiation into electrical signals using a photo-sensitive semiconductor material
- Output directly represents UV intensity in the environment

Pulse Oximeter (Active Sensor)

How to Identify

Used in scenarios involving:

- Blood oxygen (SpO₂) measurement
- Heart rate monitoring
- Finger clip medical device
- Patient monitoring systems

Working Principle

- Pulse oximeter uses:
 - Red LED
 - Infrared LED
 - Photodetector

- Oxygenated and deoxygenated hemoglobin absorb light differently

Working Mechanism

1. Light Transmission

- Red and infrared light pass through finger or ear tissue

2. Light Absorption

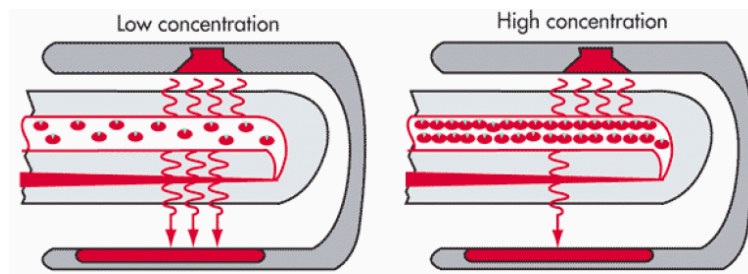
- Oxygenated blood absorbs more infrared light
- Deoxygenated blood absorbs more red light

3. Signal Detection

- Photodetector receives transmitted light
- Measures absorbed light intensity

4. Calculation

- Device calculates:
 - Blood oxygen level (SpO_2)
 - Heart rate (pulse rate)



Final Output

Displays:

- Oxygen saturation (%)

- Heart rate (BPM)

Key Idea

- Different blood conditions absorb different light wavelengths differently
- This helps measure oxygen level without drawing blood

Laser Sensor

How to Identify

Used in scenarios involving:

- Distance measurement
- Object detection
- Robotics and automation
- Positioning and alignment systems

Working Mechanism

1. Laser Emission

- Sensor sends a laser beam toward an object

2. Reflection

- Laser light reflects from the object
- Receiver lens collects the reflected light

3. Light Detection

- Reflected light falls on a photo detector
- An image of the light spot is formed

4. Distance Change

- If object distance changes:
 - Reflection angle changes
 - Position of light spot also changes

5. Distance Calculation

- Sensor measures the position shift of the light spot
- Calculates object distance accurately

Key Idea

- Laser sensor works using:
 - Laser reflection
 - Angle change
 - Light spot position shift

Output

- Accurate distance or position signal

Example

- Industrial robots
- Automatic door systems
- Object positioning machines

Magnetic & Electromagnetic Sensors

Magnetic and electromagnetic sensors are used to detect magnetic fields, movement of magnets, or changes in electromagnetic conditions. They are widely used in position sensing, speed detection, and switching applications.

1. Reed Switch

How to Identify

A reed switch is identified in scenarios involving:

- Door/window security systems
- Magnetic contact switches
- Proximity detection using magnets



Working Mechanism

- A reed switch has two thin metal strips (reed contacts)
- These are sealed inside a glass tube

Normal Condition

- No magnetic field
- Contacts stay open (OFF state)

When Magnet is Near

- Magnetic field magnetizes the metal strips
- Strips attract each other
- Contacts touch and close
- Circuit becomes ON

When Magnet is Removed

- Magnetic force disappears
- Strips separate again
- Circuit returns to OFF state

Output

- Simple ON/OFF switching signal

2. Hall Effect Sensor

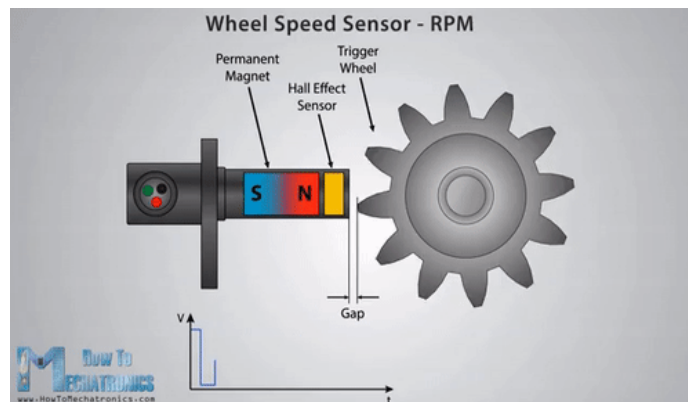
How to Identify

A Hall-effect sensor is identified in scenarios involving:

- Speed measurement
- Position sensing
- Wheel rotation detection
- Magnetic field detection

Working Mechanism

- Works based on the Hall effect principle
- When a current-carrying semiconductor is placed in a magnetic field:
 - A voltage is generated perpendicular to current flow (Hall voltage)
- Strength of this voltage depends on:
 - Strength of magnetic field
- When magnetic field changes:
 - Output voltage changes proportionally



Output

- Analog or digital voltage signal depending on magnetic field strength

Example Applications

- Speed sensors in vehicles
- Brushless DC motor control
- Position sensing systems

3. Electromagnetic Sensor

How to Identify

Used in scenarios involving:

- Metal detection
- Inductive sensing
- Industrial proximity sensing
- Magnetic field or current detection

Working Mechanism

- Works based on electromagnetic induction principle
- Sensor coil produces an alternating magnetic field

When Metal Object is Near

- Metal enters the magnetic field
- Eddy (small circular electric currents) currents are produced in the metal
- These currents disturb the original magnetic field
- Change in magnetic field is detected by the coil
- This change is converted into an electrical signal

Output

- Electrical signal depends on:
 - Presence of metal object
 - Distance of object
 - Change in magnetic field

Example

- Metal detectors
- Industrial proximity sensors
- Traffic sensing systems

Summary

- Reed switch → magnet physically closes contacts (ON/OFF)
- Hall-effect sensor → measures magnetic field as voltage change
- Electromagnetic sensor → detects changes in magnetic field using induction

Difference

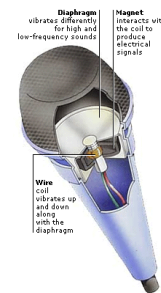
- Reed switch → mechanical magnetic switching
- Hall sensor → electrical magnetic sensing
- EM sensor → induction-based field disturbance sensing

Sound-Based Sensor

i. Microphone

Working Mechanism of Microphone

- A microphone is a sound transducer that converts sound waves (mechanical vibrations) into electrical signals.
- It works using a piezoelectric material or electromagnetic principle, depending on the type of microphone.



Piezoelectric Microphone Mechanism

What is Piezoelectric Material?

- Piezoelectric material is a special type of crystal or ceramic that generates electric charge when mechanical stress or vibration is applied.
- It is used because it can directly convert mechanical energy → electrical energy without external complex conversion systems.

Why Piezoelectric Material is Used

- It directly responds to sound-induced vibrations
- It is highly sensitive to small pressure changes
- It produces a measurable electrical signal without needing complex circuitry

Working Principle

- Sound waves are mechanical pressure waves in air
- When sound waves hit the microphone diaphragm:
 - The diaphragm starts vibrating
- These vibrations are transferred to the piezoelectric crystal
- Due to vibration:
 - The crystal molecules get mechanically stressed and displaced
 - This causes movement in charge distribution inside the material
 - As a result, micro-level electrical charges (microelectricity) are generated

Signal Conversion

- Mechanical vibration → crystal deformation
- Crystal deformation → charge imbalance
- Charge imbalance → small electrical voltage (microelectric signal)
- This electrical signal is proportional to sound intensity

Output

- A weak electrical signal (analog signal) representing sound
- This signal is later amplified and processed into audio output

Summary

- Microphone converts sound energy into electrical energy
- Piezoelectric material is used because it generates electricity from mechanical vibration

- Sound waves vibrate the material → produces microelectric signals → electrical output

ii. Ultrasonic Proximity Sensor

How to Identify

An ultrasonic proximity sensor is identified in scenarios involving:

- Distance measurement
- Obstacle detection (robots, cars, parking sensors)
- Level detection (water/liquid level)
- Short-range object detection

Working Principle

- An ultrasonic sensor works using sound waves with frequency above human hearing range (>20 kHz)
- It uses the principle of echo and time-of-flight measurement

Working Mechanism

- The sensor has two main parts:
 - Transmitter (ultrasonic transmitter)
 - Receiver (ultrasonic receiver)
- The transmitter emits high-frequency ultrasonic sound waves toward an object
- These waves travel through air and hit the object
- The waves are then reflected back as an echo
- The receiver detects the reflected waves
- The sensor measures the time taken between sending and receiving the signal (Time of Flight)

Distance Calculation Concept

- Distance is calculated using:

- Speed of sound in air
- Time taken for echo return
- Formula idea:
 - $\text{Distance} \propto (\text{Time taken} \times \text{Speed of sound}) / 2$
- Division by 2 is used because sound travels:
 - Sensor → object → back to sensor

Output

- A distance value (usually in cm or meters)
- Can also trigger ON/OFF output if object is within threshold range

Summary

- Ultrasonic sensor uses sound waves (not light)
- Works on echo reflection principle
- Measures distance using time difference between transmission and reception
- Widely used for obstacle detection and proximity sensing systems

Temperature Sensors (RTD, Thermocouple, Thermistor)

Temperature sensors are devices used to measure temperature by converting temperature changes into electrical signals such as resistance or voltage changes.

1. Resistance Temperature Detector (RTD)

- A Resistance Temperature Detector (RTD) is also called a Resistance Thermometer
- It is used to measure temperature by measuring the resistance of a metal wire
- This wire acts as a temperature sensing element

How to Identify

RTD is identified in scenarios involving:

- High accuracy temperature measurement
- Industrial temperature monitoring
- Precision sensing systems

Working Principle

- RTD works on the principle that resistance of metals changes with temperature
- It shows a nearly linear relationship between temperature and resistance
- When temperature increases:
 - Resistance of metal wire increases
- When temperature decreases:
 - Resistance decreases

Materials Used

- Common metals used:
 - Platinum
 - Nickel
 - Copper

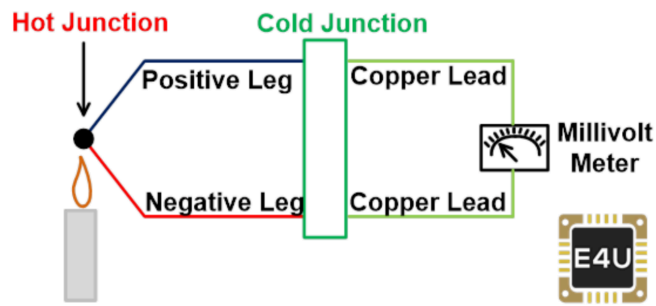
Each metal has different resistance-temperature characteristics.

Key Feature

- Very high accuracy
- Very stable and linear response
- Suitable for wide temperature range

2. Thermocouple

- A thermocouple is a temperature sensor based on thermoelectric effect
- It converts temperature difference into electrical voltage



How to Identify

Thermocouple is identified in scenarios involving:

- High temperature measurement
- Industrial furnaces
- Wide temperature range sensing

Structure

- Made of two different metal plates
- Joined at one end forming a junction

Junction Types:

- Hot junction → placed where temperature is measured
- Cold junction (reference junction) → kept at room temperature

Working Principle

- When there is a temperature difference between two junctions:
 - A small voltage is generated
- This is due to the thermoelectric effect
- The voltage depends on temperature difference
- A voltmeter measures this voltage

Output

- Voltage signal proportional to temperature difference

- Temperature is calculated from voltage value

Key Features

- Simple and robust
- Low cost
- Works in very high temperature ranges

3. Thermistor

- A thermistor is a type of resistor whose resistance changes with temperature
- It is a passive temperature sensor

How to Identify

Thermistor is identified in scenarios involving:

- Digital thermometers
- Household temperature control
- Circuit protection systems

Working Principle

- Thermistor resistance depends on temperature changes
- Resistance is measured using an ohmmeter or circuit
- By knowing the resistance-temperature relationship:
 - Temperature can be calculated

Types of Thermistor

1. NTC (Negative Temperature Coefficient)

- Temperature increases → resistance decreases

- Temperature decreases → resistance increases
- Inverse relationship

2. PTC (Positive Temperature Coefficient)

- Temperature increases → resistance increases
- Temperature decreases → resistance decreases
- Direct relationship

Materials Used

- Made using polymer or ceramic materials

Applications of Thermistor

- Digital thermometers
- Automotive temperature sensors (engine oil, coolant)
- Household appliances (fridge, oven, microwave)
- Circuit protection (current limiting)
- Rechargeable battery temperature control
- Thermal conductivity measurement
- Wheatstone bridge circuits
- Temperature compensation in circuits

Key Feature

- Very sensitive to small temperature changes
- Non-linear response

Comparison

Sensor	Output Type	Principle
RTD	Resistance	Metal resistance increases with temperature

Thermocouple	Voltage	Thermoelectric effect (temperature difference)
Thermistor	Resistance	Semiconductor resistance changes with temperature

- RTD → accurate resistance-based temperature sensor
- Thermocouple → voltage-based temperature difference sensor
- Thermistor → highly sensitive resistance-based temperature sensor

Proximity Sensor

How to Identify

A proximity sensor is identified in scenarios involving:

- Mobile phone screen turning OFF during calls
- Ear detection in smartphones
- Object detection without contact

Working Mechanism

- It consists of an infrared (IR) LED and an IR detector
- The IR LED emits infrared light continuously
- When an object (like a human ear) comes near:
 - IR light reflects back to the detector
- The detector senses the reflected IR light
- Based on intensity of reflection:
 - Sensor determines distance/close presence of object

Output

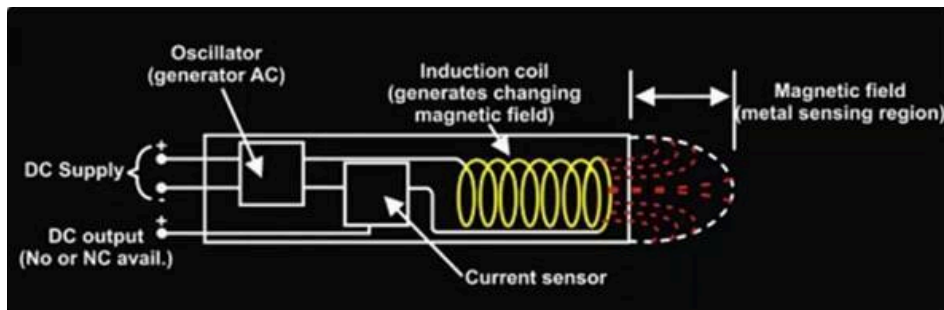
- Detects near or far object presence
- Used to turn OFF display and disable touch during calls

i. Inductive Proximity Sensor

How to Identify

It is identified in scenarios involving:

- Metal detection without contact
- Industrial automation
- Machine safety systems



Working Mechanism

- Contains a coil that generates a high-frequency magnetic field
- When a metal object approaches:
 - Eddy currents are induced in the metal due to electromagnetic induction
- These eddy currents:
 - Increase load on oscillator circuit
 - Reduce or stop oscillation
- Sensor detects this change using an amplitude detection circuit
- Output signal is generated

Output

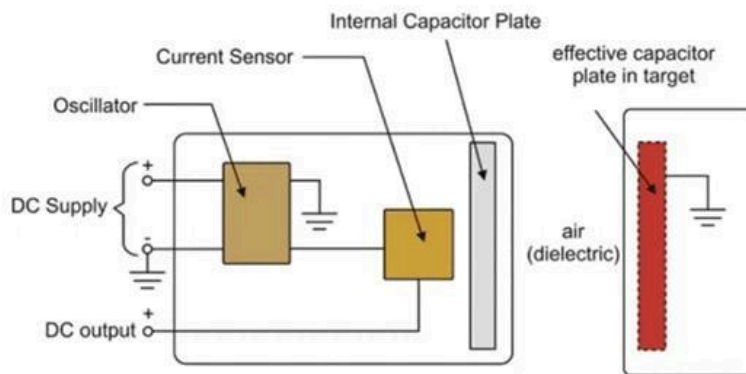
- ON/OFF signal indicating metal presence

ii.Capacitive Sensor

How to Identify

Used in scenarios involving:

- Touch sensing
- Liquid level detection
- Material detection (plastic, liquid, etc.)
- Thickness or position sensing



Working Mechanism

- Works on the principle of capacitance change
- Sensor acts like a capacitor
- Has:
 - Internal metal plate (sensor electrode)
 - External object acts as second plate
- It generates an electrostatic field
- When an object approaches:
 - Capacitance increases
 - Oscillation changes
- When threshold is reached:
 - Output is activated

Output

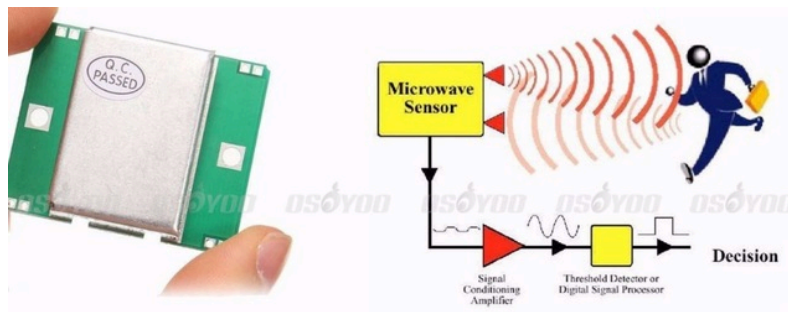
- Detects presence based on change in capacitance

iii. Microwave Motion Sensor (Radar / Doppler Sensor)

How to Identify

Used in scenarios involving:

- Motion detection (even through walls in some cases)
- Security systems
- Automatic doors and alarms



Working Mechanism

- Sensor continuously emits microwave (RF) signals
- These signals reflect back from objects in the environment
- In a stable environment:
 - Reflected signals return in a fixed pattern
- When motion occurs:
 - Object changes signal path
 - Reflected signal timing changes
- Sensor detects this change in frequency/time pattern
- If change is significant:
 - Motion is detected
 - Alarm or output is triggered

Key Point

- Small disturbances (like insects or leaves) usually do not trigger detection
- Only significant motion changes the signal pattern

Summary

- Proximity sensor → IR reflection-based detection (short range, smartphones)
- Inductive sensor → detects metal using electromagnetic induction
- Capacitive sensor → detects any material using capacitance change
- Microwave sensor → detects motion using RF wave reflection changes

Smoke Sensors

Smoke sensors are used to detect the presence of smoke particles in the air and trigger an alarm for fire safety. There are two main types:

1. Ionization Smoke Sensor

How to Identify

- Used for detecting fast flaming fires
- Found in fire alarms in homes and buildings



Working Mechanism

- Contains a small amount of radioactive material (e.g., Americium-241) placed between two electrically charged plates
- This radioactive source ionizes the surrounding air
- Ionization creates charged particles (ions)
- These ions allow a small current to flow between the plates
- When smoke enters the chamber:
 - Smoke particles attach to ions
 - Ion flow is disturbed
 - Current flow decreases
- The reduction in current is detected by the circuit
- This triggers the alarm signal

Output

- Alarm is activated when current drops due to smoke interference

2. Photoelectric Smoke Sensor

How to Identify

- More responsive to smoldering (slow-burning) fires
- Used in modern smoke detectors

Working Mechanism

- A light source (LED) is placed inside a sensing chamber
- A light sensor (photodetector) is placed at an angle (not directly aligned)
- Under normal conditions:
 - Light does not directly hit the sensor
 - No alarm is triggered
- When smoke enters:
 - Smoke particles scatter (reflect) the light
 - Reflected light reaches the sensor
 - Light intensity at sensor increases
- The sensor detects this change in light
- Alarm is triggered

Output

- Alarm activated based on scattered light detection

Key Difference (Smoke Sensors)

- Ionization → detects fast flaming fire (ion disruption method)
- Photoelectric → detects smoldering fire (light scattering method)

Gas Sensor (Chemiresistor Type)

How to Identify

- Gas leakage detection systems
- Air quality monitoring
- Industrial safety systems

Working Mechanism

- Most gas sensors use Tin Dioxide (SnO_2) as sensing material
- SnO_2 is an n-type semiconductor

In Normal Air

- Oxygen molecules are present in air
- Oxygen captures free electrons from SnO_2
- Electrons get trapped on surface
- No free electrons available
- No current flow (or very low conductivity)

In Presence of Gas (Toxic/Combustible)

- Gas reacts with adsorbed oxygen
- Oxygen is removed from surface
- Free electrons are released back
- Conductivity increases
- More gas concentration → more free electrons → higher current

Module Structure

Gas sensor module includes:

- Vcc (Power supply)
- GND (Ground)
- Digital output (0 or 1) → gas detected or not
- Analog output → gas concentration level

Signal Processing

- Output is very small initially (mV range)
- Uses LM393 comparator circuit
- Potentiometer sets threshold level
- If gas exceeds threshold → digital output becomes HIGH

Output

- Detects presence and concentration of gas

Humidity Sensor (DHT22)

How to Identify

- Used in weather monitoring systems
- Temperature and humidity measurement
- Smart home systems

Structure

DHT22 contains:

- Humidity sensing component
- NTC thermistor (temperature sensor)
- Processing IC

Humidity Working Mechanism

- Humidity sensing element has:
 - Two electrodes
 - Moisture-absorbing material between them
- When humidity changes:
 - Moisture content changes
 - Conductivity/resistance between electrodes changes

- This change is measured and processed by IC
- Output is sent to microcontroller

Temperature Working Mechanism

- Uses NTC thermistor (Negative Temperature Coefficient)
- Thermistor is a resistor whose resistance changes with temperature
- When temperature increases:
 - Resistance decreases
- When temperature decreases:
 - Resistance increases
- IC processes this change and gives temperature output

Output

- Digital data for:
 - Temperature
 - Humidity

Note

- Smoke sensor → detects smoke using ionization or light scattering
- Gas sensor → detects gas using chemical reaction and conductivity change
- Humidity sensor → measures moisture using resistance change
- Temperature sensor (NTC) → measures temperature via resistance variation

Touch Sensor

Touch sensors work like an electronic switch. When touch, pressure, or force is applied, they get activated and behave like a closed switch (ON state). When the touch is removed, they return to an open switch (OFF state).

1. Capacitive Touch Sensor

How to Identify

- Used in smartphones, touch screens, iPods
- No need of physical pressure
- Works with finger touch only

Structure

- Contains two parallel conductive plates
- Separated by an insulating material
- Forms a capacitor with a fixed capacitance value

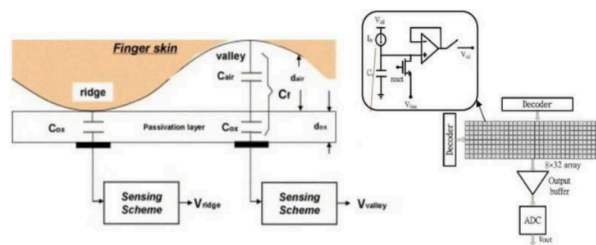
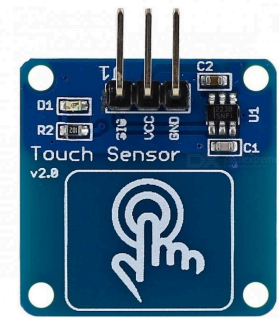


Image is showing How a capacitive fingerprint scanner captures digital images (Source: Android Authority/Robert Triggs)

Working Mechanism

- The sensor continuously monitors capacitance value using a measuring circuit
- Under normal condition:
 - Capacitance remains stable
- When a finger touches the surface:
 - Human finger acts as a conductive object
 - It disturbs the electric field
 - Causes an increase in capacitance value
- The sensing circuit detects this change
- A signal is generated indicating touch detection

Output

- Electrical signal when capacitance changes due to touch

Key Point

- Works on capacitance change, not pressure

Limitation

- Can sometimes give false triggering

2. Resistive Touch Sensor

How to Identify

- Used in:
 - Keypads
 - Touch pads
 - Musical instruments
- Requires physical pressure

Structure

- Made of two conductive layers (films)
- Coated with Indium Tin Oxide (ITO) (good conductor)
- Layers are separated by a small gap

Working Mechanism

- A constant voltage is applied across the conductive layers
- When no touch:
 - Layers remain separated
 - No signal change

- When pressure is applied:
 - Top layer touches bottom layer
 - Electrical contact is formed
 - A voltage drop is created
- Controller detects this voltage change
- Touch is registered

Output

- Electrical signal based on pressure-induced voltage change

Key Point

- Works on physical pressure and contact

Limitation

- Requires strong pressure to operate

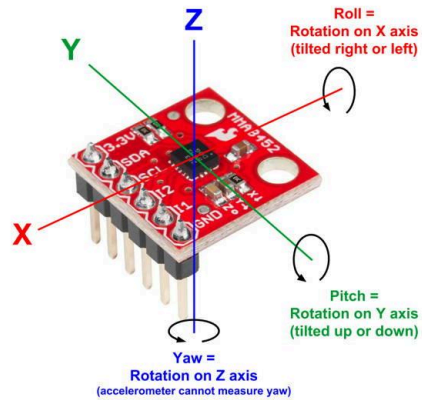
Note

- Capacitive sensor → detects change in capacitance caused by finger touch
- Resistive sensor → detects physical pressure causing voltage change
- Both act like electronic switches (ON/OFF detection)

Accelerometer

How to Identify

- Motion sensing in phones, gaming devices, drones
- Orientation, tilt, vibration, and movement detection



Working Principle

- An accelerometer measures acceleration due to motion and gravity
- It helps determine:
 - Device tilt (orientation with respect to Earth)
 - Speed and direction of movement

Working Mechanism

- It works using two main principles:
 - Piezoelectric effect
 - Capacitive sensing

Piezoelectric Type

- Contains microscopic crystal structures (piezoelectric material)
- When acceleration or vibration occurs:
 - The mass inside the sensor applies force on the crystal
 - Crystal gets mechanically stressed (compressed/squeezed)
 - This generates an electrical charge (voltage)
- The generated voltage is:
 - Proportional to applied force
 - Hence proportional to acceleration

Capacitive Type

- Uses microstructures with capacitance sensors
- When device moves:
 - Position of internal mass changes
 - Distance between capacitor plates changes
 - This changes capacitance value
- Circuit converts capacitance change → voltage signal

Output

- Electrical signal representing:
 - Acceleration
 - Tilt angle
 - Motion direction

Gyroscope Sensor

How to Identify

- Used in:
 - Smartphones (screen rotation)
 - Drones, navigation systems
 - Robotics

Working Principle

- Measures angular velocity (rotation speed)
- Works using Coriolis force principle

Working Mechanism

- Inside sensor, a vibrating structure (drive arm) is present
- Normally, it vibrates in a fixed direction

When device rotates:

- Coriolis force acts on vibrating structure
- Causes secondary vibration in sensing direction
- This creates a motion difference in sensing arms
- Movement produces voltage difference
- Voltage is proportional to angular velocity

Output

- Electrical signal representing:
 - Rotation speed
 - Angular movement

Door / Elevator Sensor

How to Identify

- Automatic doors
- Elevator safety systems
- Obstacle detection in doorway

Working Principle

- Uses infrared beam system (TX and RX pair)
- Works as a safety detection system

Working Mechanism

- Transmitter (TX) emits infrared beam
- Receiver (RX) receives the beam

Normal Condition:

- Beam is continuous → door closes normally

Obstruction Condition:

- Person/object blocks IR beam
- Receiver does not receive signal
- System detects obstruction

Response:

- Door reopens automatically
- Remains open until obstruction is removed
- If delay continues:
 - System enters nudge mode (slow closing with alarm beep)

Output

- ON/OFF signal for door control system

Fingerprint Sensor

How to Identify

- Mobile phone unlock
- Security systems
- Attendance systems

Types

- Optical fingerprint sensor
- Capacitive fingerprint sensor

Optical Sensor

Working Mechanism

- Finger is illuminated using light source
- Camera captures fingerprint image

- Image is processed into digital pattern
- Ridges and valleys converted into binary data (0s and 1s)

Capacitive Sensor

Working Mechanism

- Uses capacitor array and human conductivity
- Finger placed on sensor:
 - Ridges touch sensor → change in capacitance
 - Valleys do not touch → no change
- Circuit detects capacitance pattern
- Converts into digital fingerprint map

Output

- Digital fingerprint pattern for authentication

LiDAR Sensor

How to Identify

- Autonomous vehicles
- 3D mapping systems
- Robotics and drones
- Distance measurement systems

Working Principle

- Uses laser light pulses and time-of-flight measurement

Working Mechanism

- Sensor emits rapid laser pulses (~150,000 per second)

- Laser hits object and reflects back
- Sensor measures time taken for return (time of flight)

Distance Calculation

- Distance is calculated using:
 - Time of flight of laser pulse
 - Speed of light
- Formula:
 - $\text{Distance} = (\text{Speed of Light} \times \text{Time}) / 2$
- Division by 2 is because:
 - Light travels to object and back

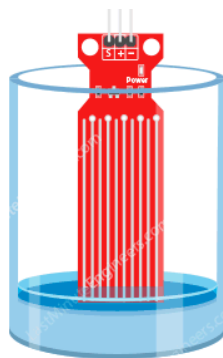
Output

- Produces accurate distance and 3D surface map

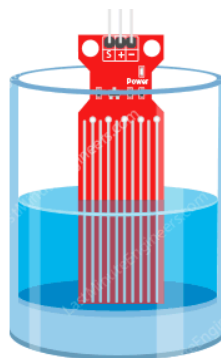
Water Level Sensor (Capacitive Type)

How to Identify

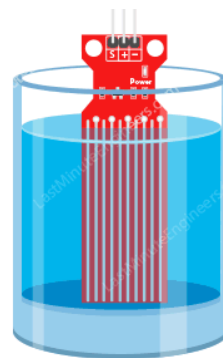
- Tank level monitoring
- Liquid detection systems
- Industrial fluid control



Status: Dry
Test Reading: ~0



Status: Partially submerged
Test Reading: ~420



Status: Fully submerged
Test Reading: ~520

Working Principle

- Based on change in capacitance due to liquid level

Working Mechanism

- Sensor has:
 - One electrode (insulated plate)
 - Tank wall acts as second electrode
- Capacitance depends on:
 - Dielectric constant of material
 - Distance between plates
 - Area of plates

Operation

Empty Tank:

- Low dielectric → low capacitance

Filled Tank:

- Liquid (e.g., water) increases dielectric constant
- Capacitance increases

Output

- Electrical signal proportional to liquid level

Note

- Accelerometer → measures linear motion and tilt
- Gyroscope → measures rotation (angular velocity)
- Door sensor → IR beam interruption detection
- Fingerprint sensor → optical or capacitive pattern recognition
- LiDAR → laser time-of-flight distance measurement
- Water level sensor → capacitance change due to liquid

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